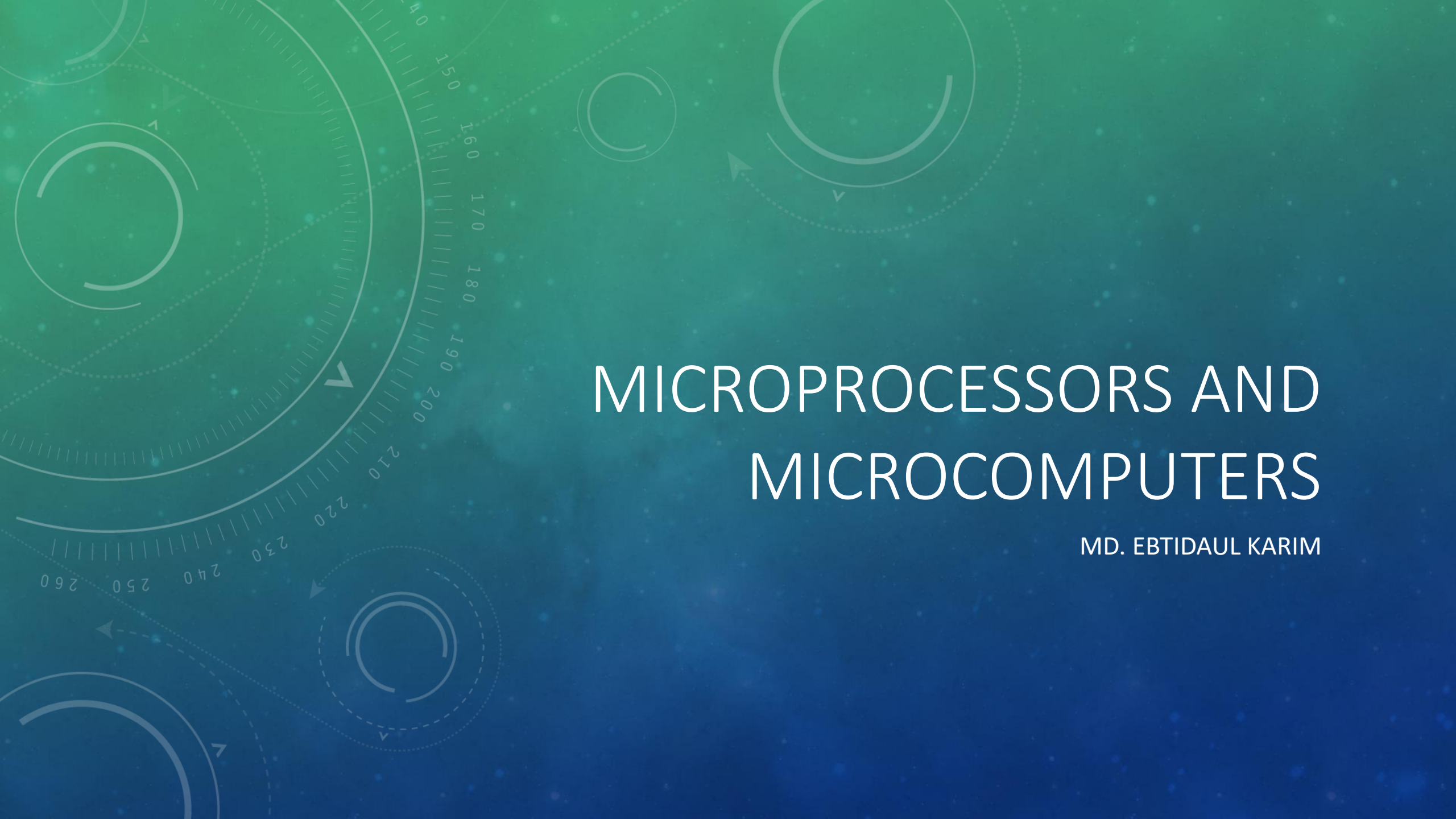


The background features a gradient from light green at the top to dark blue at the bottom. On the left side, there are several concentric circular patterns and a large arc with a scale ranging from 140 to 260. The scale is marked with numbers every 10 units. The overall design is technical and futuristic.

MICROPROCESSORS AND MICROCOMPUTERS

MD. EBTIDAUl KARIM

HISTORY

- On December 23, 1947 at Bell Labs by John Bardeen, William Shockley, and Walter Brattain invented transistor. It was a major breakthrough.
- Fairchild Semiconductors (founded in 1957) invented the first IC in 1959.
- In 1968, Robert Noyce, Gordon Moore, Andrew Grove resigned from Fairchild Semiconductors.
- They founded their own company Intel (Integrated Electronics).
- The world's first microprocessor, the Intel 4004, was a 4-bit microprocessor-programmable controller on a chip.

MICROPROCESSORS

- Microprocessor is a controlling unit of a micro-computer, fabricated on a small chip capable of performing ALU (Arithmetic Logical Unit) operations and communicating with the other devices connected to it.
- The microprocessor is a multipurpose, clock driven, register based, digital integrated circuit that accepts binary data as input, processes it according to instructions stored in its memory and provides results (also in binary form) as output.
- Microprocessor consists of an ALU, register array, and a control unit.

MICROPROCESSORS

- ALU performs arithmetical and logical operations on the data received from the memory or an input device.
- Register array consists of registers identified by letters like B, C, D, E, H, L and accumulator.
- The control unit controls the flow of data and instructions within the computer.

ADVANTAGES OF A MICROPROCESSOR

- **Low Cost**

Microprocessors are available at low cost due to integrated circuit technology. Which will reduce the cost of a computer system.

- **High Speed**

Microprocessor chips can work at very high speed due to the technology involved in it. It is capable of executing millions of instructions per second.

- **Small Size**

Due to very large scale and ultra large scale integration technology, a microprocessor is fabricated in a very less footprint. This will reduce the size of the entire computer system.

ADVANTAGES OF A MICROPROCESSOR

- **Versatile**

Microprocessors are very versatile, the same chip can be used for a number of applications by simply changing the program (instructions stored in the memory).

- **Low Power Consumption**

Microprocessors are usually manufactured using **metal oxide semiconductor** technology, in which MOSFETs (Metal Oxide Semiconductor Field Effect Transistors) are working in saturation and cut off modes. So the power consumption is very low compared to others.

- **Less Heat Generation**

Compared to vacuum tube devices, semiconductor devices won't emit that much heat.

ADVANTAGES OF A MICROPROCESSOR

- **Reliable**

Microprocessors are very reliable, failure rate is very less as semiconductor technology is used.

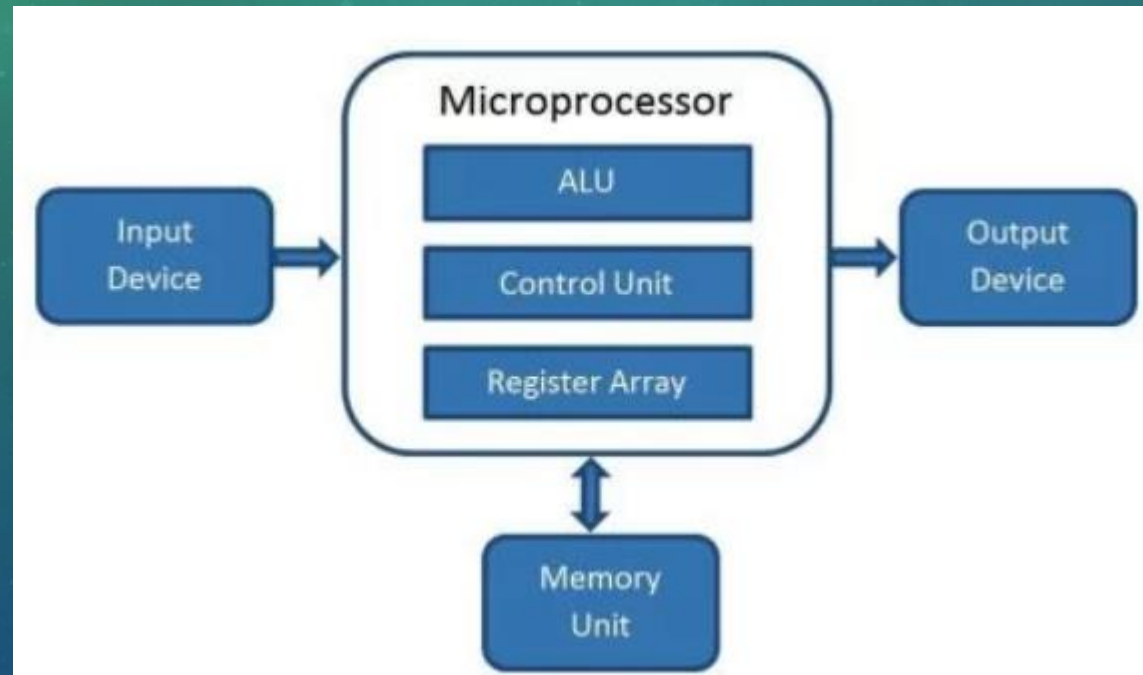
- **Portable**

Devices or computer system made with microprocessors can be made portable due to the small size and low power consumption.

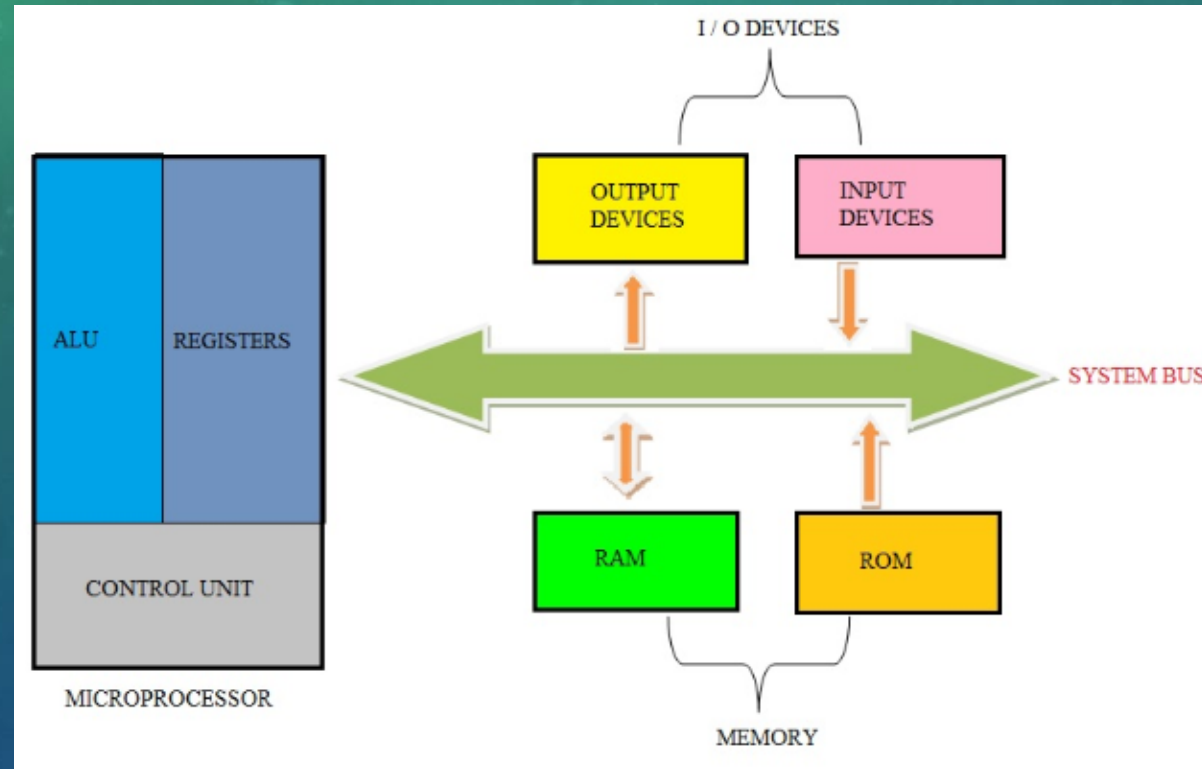
MICROCOMPUTER

- A **microcomputer** is a small, relatively inexpensive computer with a microprocessor as its central processing unit (CPU)
- The term microcomputer is not as commonly used as it was during the 1970s-1980s. We now refer to microcomputers as, simply, computers, or personal computers (PC).
- It includes a microprocessor, memory and minimal input/output (I/O) circuitry mounted on a single printed circuit board (PCB).

BLOCK DIAGRAM OF A BASIC MICROCOMPUTER



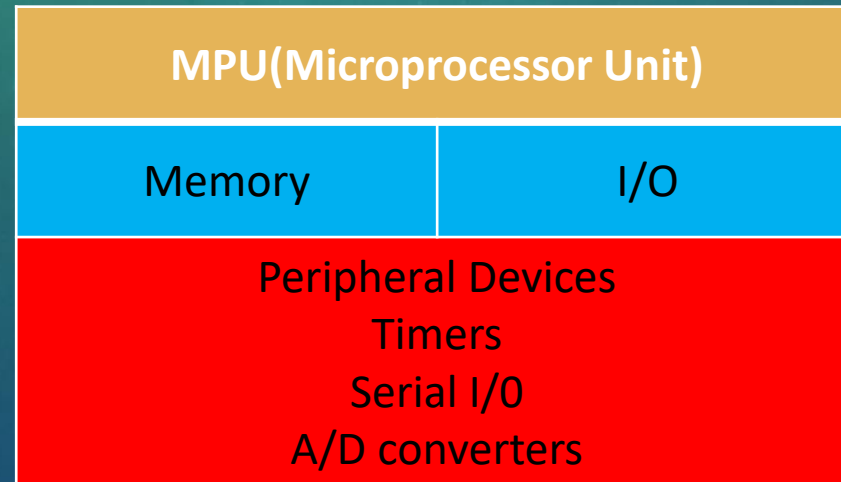
BLOCK DIAGRAM OF A BASIC MICROCOMPUTER



MICROCONTROLLER

- Even though microprocessors are powerful devices, they require external chips like RAM, ROM, Input / Output ports and other components in order to design a complete working system.
- This made it economically difficult to develop computerized consumer appliances on a large scale as the system cost is very high.
- Microcontrollers are the devices that actually fit the profile “Computer – on – a chip” as it consists of a main processing unit or processor along with some other components that are necessary to make it a complete computer.
- The components that are present on a typical microcontroller IC are CPU, memory, input / output ports and timers.

MICROCONTROLLER



COMMON TERM USED IN MICROPROCESSORS

- **Bus**

- A bus is a set of conductors intended to transmit data, address or control information to different elements in a microprocessor. Usually a microprocessor will have 3 types of buses : Data Bus, Control Bus and Address Bus. An 8-bit processor will be using 8-bit wide bus.

- **Instruction Set**

- **Instruction set is the group of commands that a microprocessor can understand.** So instruction set is an interface between hardware and software (program). An instruction commands the processor to switch relevant transistors for doing some processing in data. For eg. ADD A, B; is used to add two numbers stored in the register A and B.

COMMON TERM USED IN MICROPROCESSORS

- **Word Length**

- Word Length is the number of bits in the internal data bus of a processor or it is the number of bits a processor can process at a time. For eg. An 8-bit processor will have an 8-bit data bus, 8-bit registers and will do 8-bit processing at a time. For doing higher bits (32-bit, 16-bit) operations, it will split that into a series of 8-bit operations.

- **Cache Memory**

- Cache memory is a random access memory that is integrated into the processor. So the processor can access data in the cache memory more quickly than from a regular RAM. It is also known as CPU Memory. Cache memory is used to store data or instructions that are frequently referenced by the software or program during the operation. So it will increase the overall speed of the operation.

COMMON TERM USED IN MICROPROCESSORS

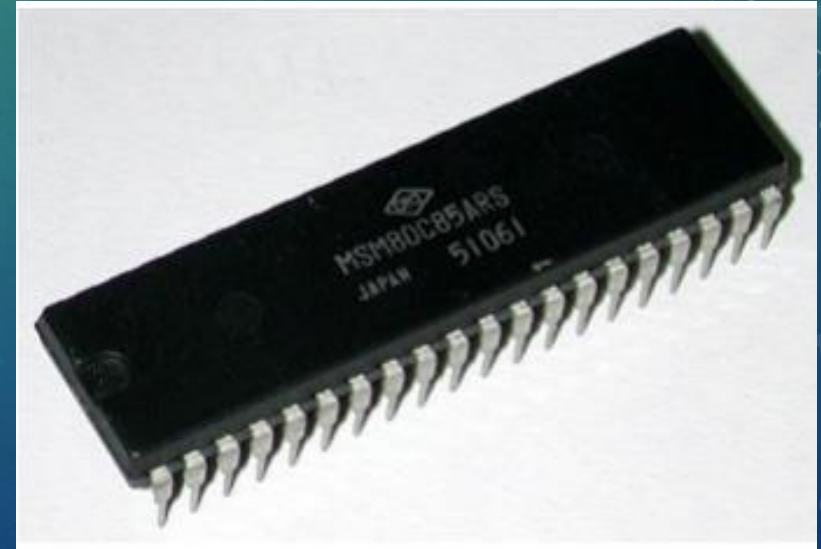
- **Clock Speed**
 - Microprocessors use a clock signal to control the rate at which instructions are executed, synchronize other internal components and to control the data transfer between them. So clock speed refers to the speed at which a microprocessor executes instructions. It is usually measured in Hertz and are expressed in megahertz (MHz), gigahertz (GHz) etc.

MICROPROCESSORS

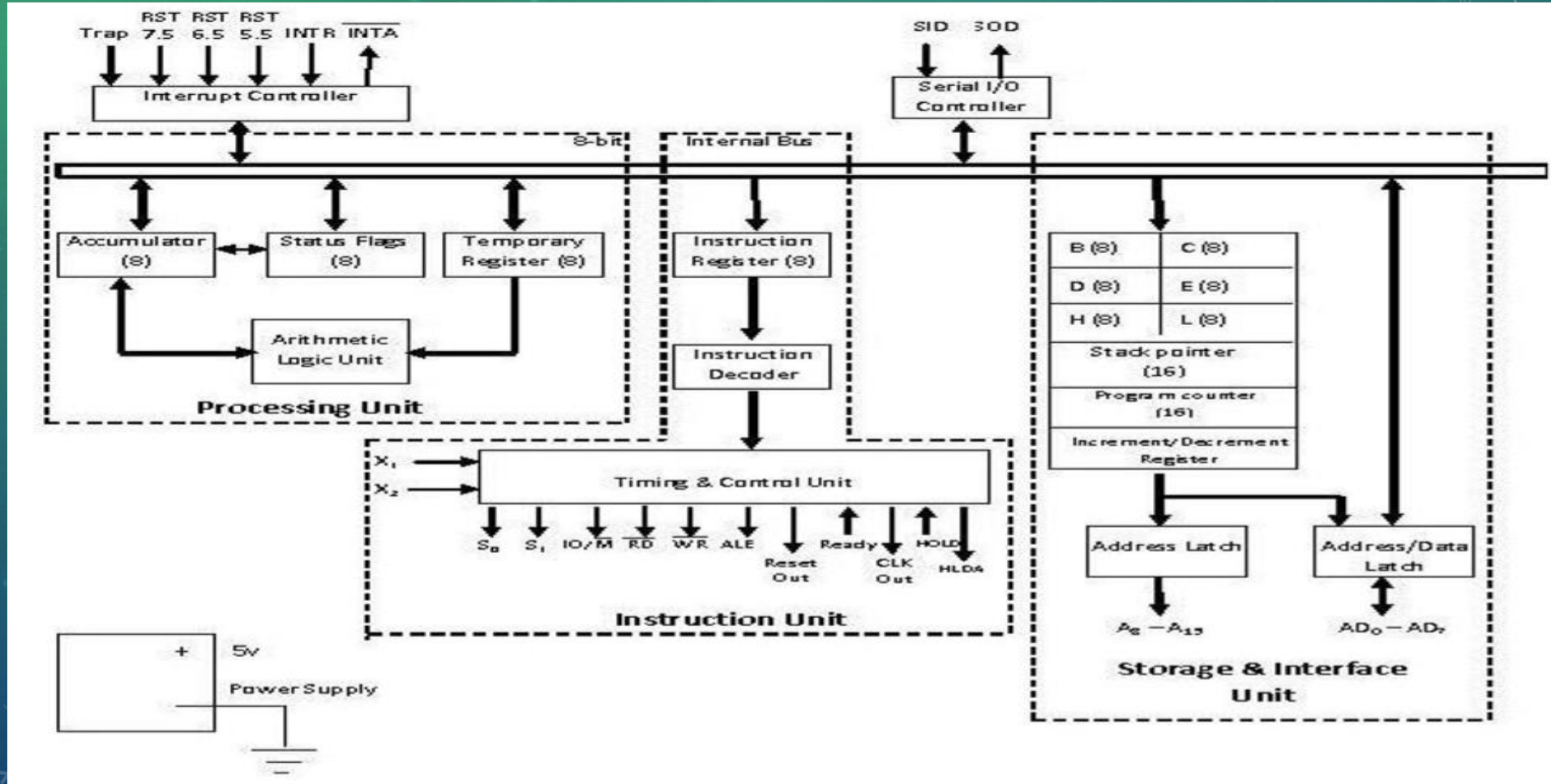
- Depending on the word length microprocessors can be categorized as:
 - i. 4-Bit Microprocessors
 - ii. 8-Bit Microprocessors
 - iii. 16-Bit Microprocessors
 - iv. 32-Bit Microprocessors
 - v. 64-Bit Microprocessors

INTRODUCTION TO 8085

- ✓ Introduced in 1977.
- ✓ It is 8-bit MP.
- ✓ It is a 40 pin dual-in-line chip.
- ✓ It uses a single +5V supply for its operations.
- ✓ Its clock speed is about 3MHz.



8085 BLOCK DIAGRAM



THREE UNITS OF 8085

- Processing Unit
- Instruction Unit
- Storage and Interface Unit

ACCUMULATOR

- It the main register of microprocessor.
- It is also called register 'A'.
- It is an 8-bit register.
- It is used in the arithmetic and logic operations.
- It always contains one of the operands on which arithmetic/logic has to be performed.
- After the arithmetic/logic operation, the contents of accumulator are replaced by the result.

ARITHMETIC & LOGIC UNIT (ALU)

- It performs various arithmetic and logic operations.
- The data is available in accumulator and temporary/general purpose registers.
- **Arithmetic Operations:**
 - ✓ Addition, Subtraction, Increment, Decrement etc.
- **Logic Operations:**
 - ✓ AND, OR, X-OR, Complement etc.

STATUS FLAGS

- Status Flags are set of flip-flops which are used to check the status of Accumulator after the operation is performed.
- It is a 8-bit register in which 5 of the bits carry significant information in the form of flags.

D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
S	Z	X*	AC	X*	P	X*	CY

X*=don't care condition

STATUS FLAGS

- S = Sign Flag
- Z = Zero Flag
- AC = Auxiliary Carry Flag
- P = Parity Flag
- CY = Carry Flag

STATUS FLAGS

❑ Sign Flag (S):

- It tells the sign of result stored in Accumulator after the operation is performed.
- If result is -ve, sign flag is set (1).
- If result is +ve, sign flag is reset (0).

STATUS FLAGS

❑ Zero Flag (Z):

- It tells whether the result stored in Accumulator is zero or not after the operation is performed.
- If result is zero, zero flag is set (1).
- If result is not zero, zero flag is reset (0).

STATUS FLAGS

❑ Auxiliary Carry Flag (AC):

- It is used in BCD operations.
- This flag is set if there is an overflow out of 3 i.e. carry from D_3 to D_4 bit (from lower nibble to higher nibble) where 4 bit= 1 nibble and 8 bit= 1 byte
- If there is no carry, auxiliary carry is reset (0).

STATUS FLAGS

❑ Parity Flag (P):

- Parity is defined by the number of ones present in the accumulator.
- If parity is even, parity flag is set (1).
- If parity is odd, parity flag is reset (0).

STATUS FLAGS

☐ Carry Flag (CY):

- This flag is set if there is an overflow out of 7.
- The carry flag also serves as a borrow flag for subtraction.

GENERAL PURPOSE REGISTERS

- There are 6 general purpose registers, namely B, C, D, E, H, L.
- Each of them is 8-bit register.
- They are used to hold data and results.
- To hold 16-bit data, combination of two 8-bit registers can be used.
- This combination is known as Register Pair.
- The valid register pairs are:
- B – C, D – E, H – L.

PROGRAM COUNTER(PC)

- It is used to hold the address of next instruction to be executed.
- It is a 16-bit register.
- The microprocessor increments the value of Program Counter after the execution of the current instruction, so that, it always points to the next instruction.

STACK POINTER(SP)

- The stack pointer is also a 16-bit register that is used to point into memory.
- The memory this register points to is a special area called the stack.
- The stack is an area of memory used to hold data that will be retrieved soon.
- The stack is usually accessed in a Last In First Out (LIFO) fashion.

INCREMENT/DECREMENT REGISTER

- This register is used to increment or decrement the value of Stack Pointer.
- During PUSH operation, the value of Stack Pointer is incremented.
- During POP operation, the value of Stack Pointer is decremented.

DATA BUS

- The data bus is 'bi-directional'
- ✓ data or instruction codes from memory or Input/output are transferred into the microprocessor
- ✓ the result of an operation or computation is sent out from the microprocessor to the memory or input/output.
- Depending on the particular microprocessor, the data bus can handle 8 bit or 16 bit data.

ADDRESS BUS

- The address bus is 'unidirectional', over which the microprocessor sends an address code to the memory or input/output.
- The size (width) of the address bus is specified by the number of bits it can handle.
- The more bits there are in the address bus, the more memory locations a microprocessor can access.
- A 16 bit address bus is capable of addressing 65,536 (64K) addresses.

CONTROL BUS

- The control bus is used by the microprocessor to send out or receive timing and control signals in order to coordinate and regulate its operation and to communicate with other devices, i.e. memory or input/output.
- Control bus is unidirectional.
- It is of 1 bit.

MACHINE LANGUAGE

- The number of bits that form the “word” of a microprocessor is fixed for that particular processor.
 - These bits define a maximum number of combinations.
 - For example **an 8-bit microprocessor can have at most $2^8 = 256$ different combinations.**
- However, in most microprocessors, not all of these combinations are used.
 - Certain patterns are chosen and assigned specific meanings.
 - Each of these patterns forms an instruction for the microprocessor.
 - The complete set of patterns makes up the microprocessor’s machine language.

THE 8085 MACHINE LANGUAGE

- The 8085 (from Intel) is an 8-bit microprocessor.
- The 8085 uses a total of 246 bit patterns to form its instruction set.
- These 246 patterns represent only 74 instructions.
- The reason for the difference is that some (actually most) instructions have multiple different formats.
- Because it is very difficult to enter the bit patterns correctly, they are usually entered in hexadecimal instead of binary.
- For example, the combination 0011 1100 which translates into “increment the number in the register called the accumulator”, is usually entered as 3C.

ASSEMBLY LANGUAGE

- Entering the instructions using hexadecimal is quite easier than entering the binary combinations.
- However, it still is difficult to understand what a program written in hexadecimal does.
- So, each company defines a symbolic code for the instructions.
- These codes are called “mnemonics”.
- The mnemonic for each instruction is usually a group of letters that suggest the operation performed.

ASSEMBLY LANGUAGE

- **Using the same example from before,**
 - ✓ 00111100 translates to 3C in hexadecimal (OPCODE)
 - ✓ Its mnemonic is: “INR A”.
 - ✓ INR stands for “increment register” and A is short for accumulator.
- **Another example is: 1000 0000,**
 - ✓ Which translates to 80 in hexadecimal.
 - ✓ Its mnemonic is “ADD B”.
 - ✓ “Add register B to the accumulator and keep the result in the accumulator”.

ASSEMBLY LANGUAGE

- So assembly language is advantageous instead of writing a program directly in machine language.
- **Self Study:**
 - ✓ Assembler.
 - ✓ Debugger.
 - ✓ Linker.

ASSEMBLY LANGUAGE

- It is important to remember that a machine language and its associated assembly language are completely **machine dependent**.
- In other words, they are not transferable from one microprocessor to a different one.
- For example, Motorola has an 8-bit microprocessor called the 6800.
- The 8085 machine language is very different from that of the 6800. So is the assembly language.
- A program written for the 8085 cannot be executed on the 6800 and vice versa.

TASK

- Following are the instruction of a 8085 microprocessor:
 - ✓ MVI A, 24H
 - ✓ MVI B, 42H
 - ✓ ADD B.
- ☐ Draw the status flag after above operation.



Thank You